



K.S.R. COLLEGE OF ENGINEERING

(AUTONOMOUS)

DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

FINAL PROJECT REVIEW

HANDMATIC: WIRELESS ROBOTIC ARM CONTROL THROUGH GESTURE RECOGNITION

Guide

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DOMAIN

❖ EMBEDDED SYSTEMS

PICO

- ❖ **Problem(P):** For industries like chemical activities and other dangerous processes, human interface leads to health issues.
- ❖ **Invention(I):** Wirelessly controlled robotic arm based on hand gestures.
- ❖ **Comparsion(C):**The current approach uses a human interface and manual control.
- ❖ **Outcome(O):**Hand gesture-based robotic arm with wireless control.

ABSTRACT

- ❖ This project presents an Arduino-based robotic arm for hazardous tasks like chemical handling and toxic material processing.
- ❖ The arm uses RF communication for wireless control, a gyroscope for orientation, and a flex sensor for precise movement.
- ❖ Fingerprint and keypad authentication ensure secure access, allowing only authorized personnel to operate it.
- ❖ Designed for remote operation in risky environments, the system combines robotics, wireless communication, and advanced sensors, enhancing safety, precision, and efficiency in industries handling dangerous materials.

INTRODUCTION

- ❖ The increasing risks of handling hazardous chemicals and working in dangerous environments have driven the development of robotic systems to ensure human safety.
- ❖ This project focuses on an Arduino-based robotic arm equipped with RF wireless control, a gyroscope for motion tracking, and a flex sensor for finger movement detection, enabling precise remote operation.
- ❖ A fingerprint recognition module and keypad authentication enhance security, ensuring only authorized personnel can operate it.
- ❖ This system minimizes human exposure to danger, providing a safe, efficient solution for **high-risk industries** like chemical processing and **toxic waste management**.

OBJECTIVE

- ❖ **Enhance Safety** – Minimize human exposure to hazardous environments by enabling remote-controlled operation of the robotic arm.
- ❖ **Improve Precision** – Utilize gyroscope and flex sensors for accurate motion tracking and precise manipulation of objects in dangerous conditions.
- ❖ **Ensure Secure Access** – Implement fingerprint authentication and keypad-based multi-factor authentication to restrict control to authorized personnel.
- ❖ **Enable Wireless Control** – Use RF communication for seamless, real-time remote operation, allowing users to control the robotic arm from a safe distance.
- ❖ **Increase Industrial Efficiency** – Provide a reliable, automated solution for handling hazardous materials in industries like chemical processing, nuclear facilities, and toxic waste management.

EXISTING SYSTEM

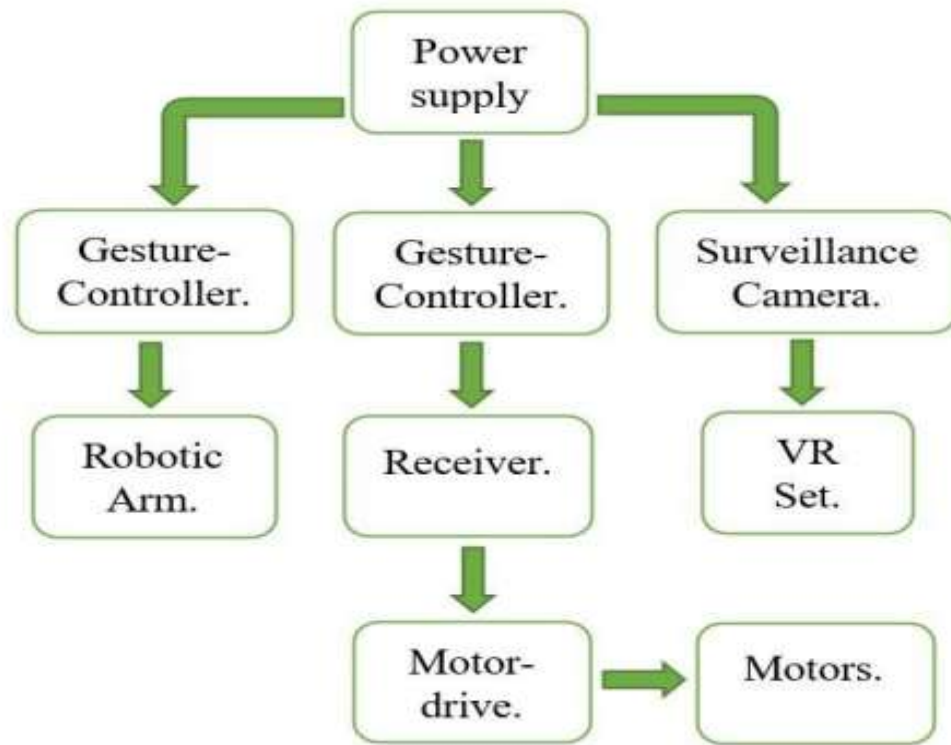


Figure 1: Block Diagram for Existing System.

- ❖ The bot's locomotion and robotic arm are controlled using a gesture-controlled transmitter and receiver, responding to hand movements.
- ❖ A glove with a gesture controller allows intuitive control of the robotic arm, mimicking human hand joints for easy operation.
- ❖ The receiver processes hand movement data and sends commands to the motor driver, which controls motors for movement.
- ❖ A compact, high-resolution FPV camera with a built-in transmitter is used for real-time monitoring.
- ❖ A pan-tilt mechanism with a servo motor syncs the FPV camera to a VR set, aligning the camera with the user's head movements.

DISADVANTAGES OF EXISTING SYSTEM

- ❖ Gesture control relies on wireless communication, which can be affected by interference, leading to delayed or inaccurate responses.
- ❖ The effectiveness of gesture control and FPV transmission depends on the range, restricting the bot's operational distance.
- ❖ The robotic system, including motors, servos, and wireless components, requires significant power, reducing battery life.
- ❖ Proper synchronization of hand gestures with the robotic arm requires precise calibration, which may be time-consuming.
- ❖ The FPV camera's pan-tilt mechanism depends on a VR set, which might not be universally compatible or comfortable for all users.

PROJECT EXPLANATION

- ❖ **Gesture-Based Control:** The robotic arm is controlled using hand gestures, detected by a gyroscope (for wrist movement) and flex sensors (for gripper control), ensuring intuitive and precise operation.
- ❖ **Wireless Communication:** The system operates wirelessly using RF modules, with a transmitter side (wearable controller) sending signals to the receiver side (robotic arm) for seamless remote operation.
- ❖ **Security Features:** Fingerprint authentication and a keypad sensor are integrated to restrict unauthorized access, ensuring secure control of the robotic arm in sensitive environments.
- ❖ **Real-Time Feedback:** An LCD display provides live updates on executed commands, authentication status, and sensor data, improving user interaction and monitoring.
- ❖ **Versatile and Safe Application:** Designed for hazardous industries, healthcare, and assistive robotics, the system enhances safety, precision, and efficiency in environments where human intervention is risky.

PROPOSED SYSTEM

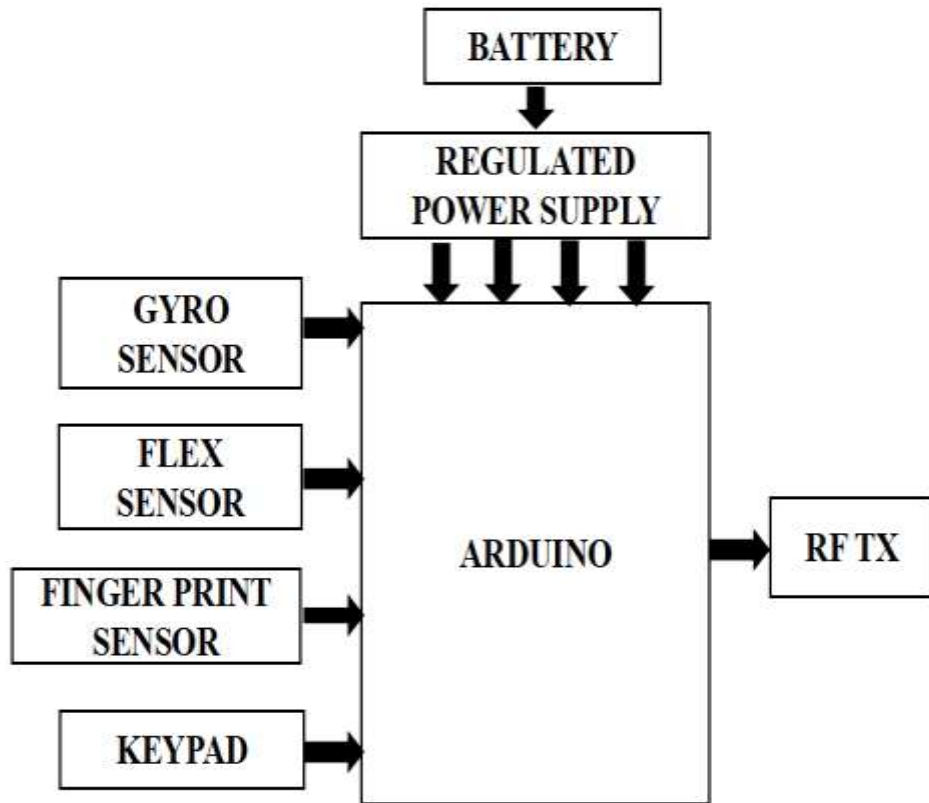


Figure 2 : Block Diagram for Proposed Transmitter System.

- ❖ The Arduino serves as the central controller, processing inputs from various sensors and devices to manage motor operation via RF communication.
- ❖ Inputs include a gyro sensor for detecting orientation or movement, a fingerprint sensor for authentication, and a keypad for user commands.
- ❖ The Arduino transmits processed commands through an RF transmitter to a remote receiver, which could be another Arduino or a dedicated controller.
- ❖ RF communication enables wireless operation, enhancing system flexibility and ensuring motor actions are based on valid sensor readings or user inputs.
- ❖ A battery powers the entire system, providing portability and standalone functionality.

PROPOSED SYSTEM

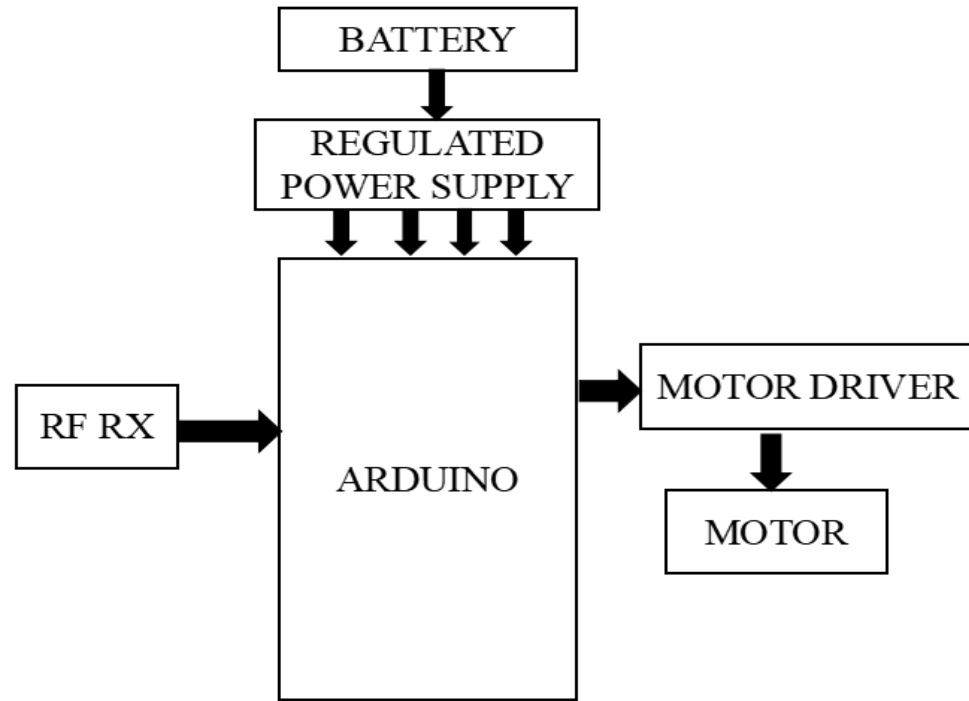


Figure 3: Block Diagram for Proposed Receiver System.

- ❖ The RF receiver collects transmitted data and sends it to the motor driver, which controls the motor based on received commands.
- ❖ The motor driver interprets signals for start/stop, speed, and direction, enabling precise motor control.
- ❖ The system is applicable in robotics, remote-controlled devices, and security systems where accuracy and authentication are essential.
- ❖ A battery-powered design ensures portability and independence from fixed power sources.
- ❖ The setup is ideal for mobile and outdoor applications due to its wireless and standalone functionality.

CIRCUIT DIAGRAM OF PROPOSED SYSTEM

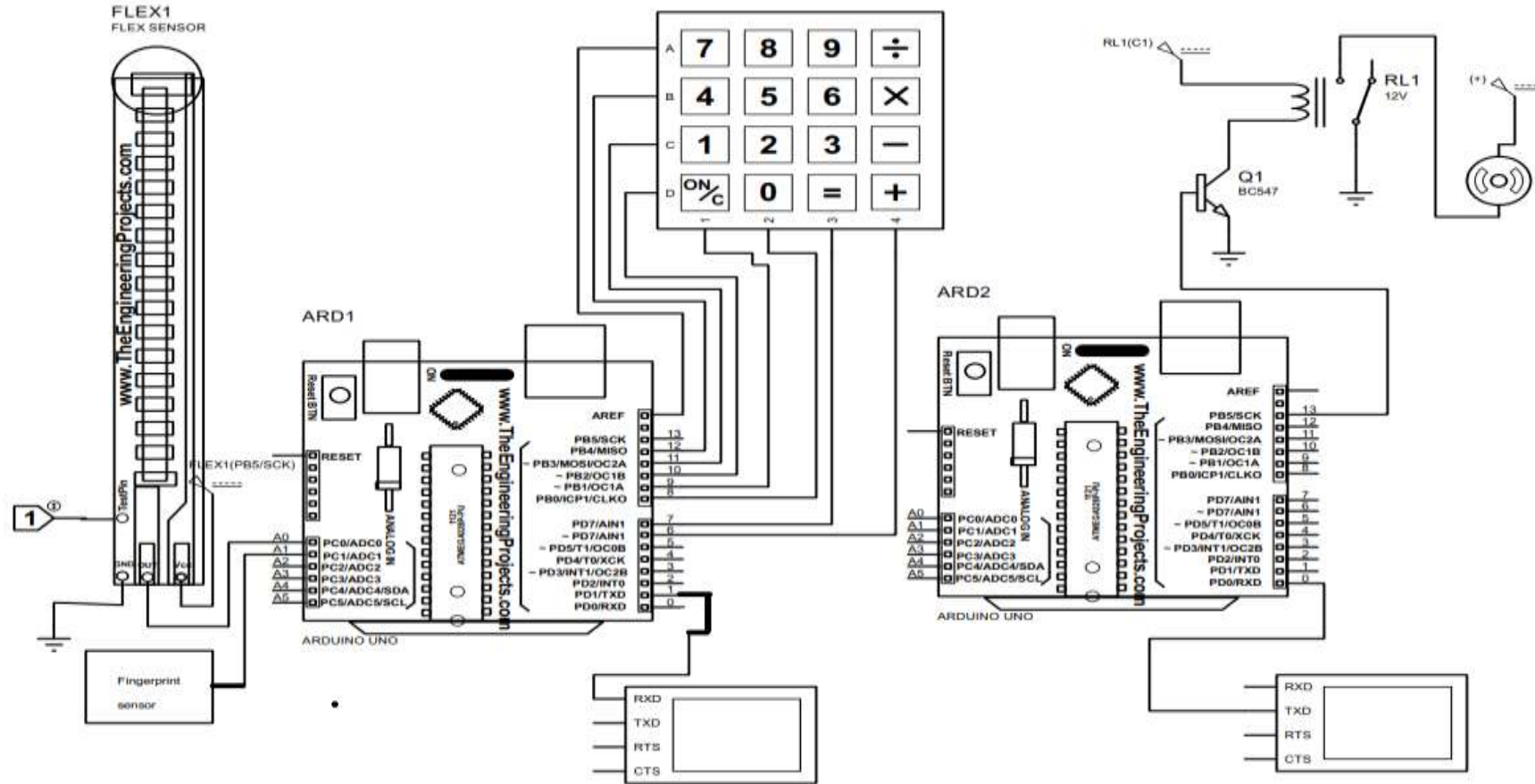
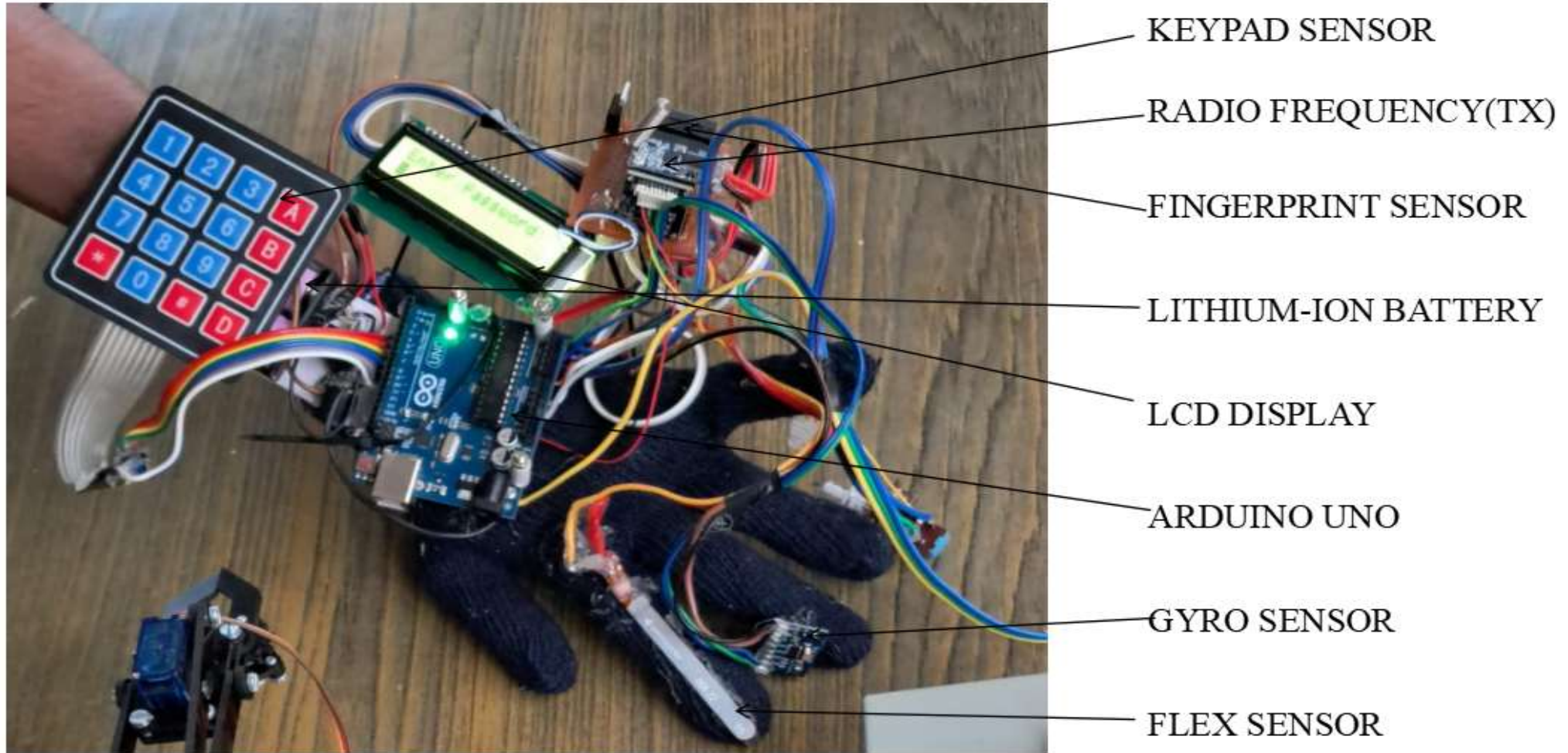


Figure 4: Circuit Diagram of Proposed System.

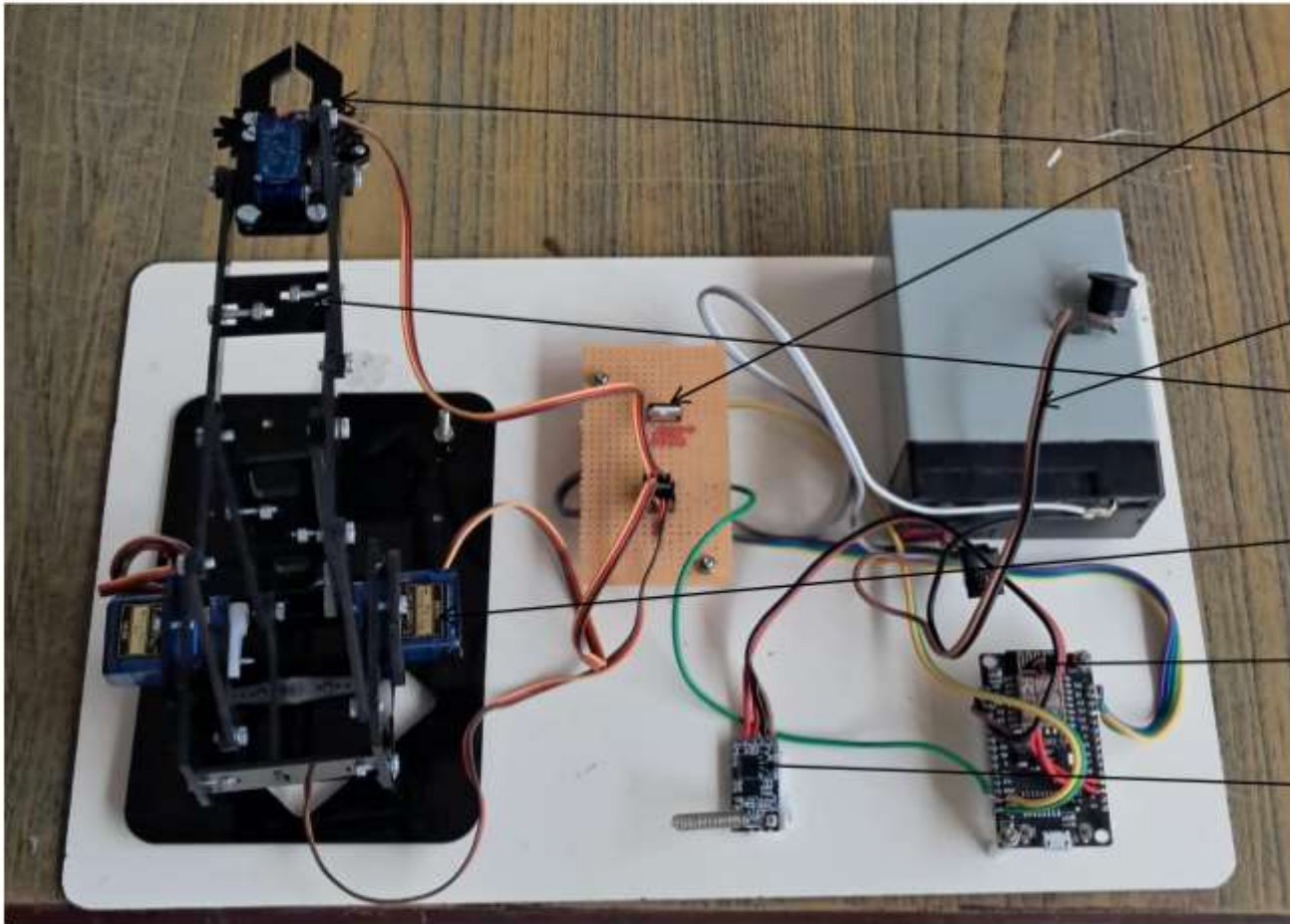
CIRCUIT DIAGRAM EXPLANATION

- The system gathers input from different sources, including sensors and user interfaces.
- The central controller processes the inputs and makes decisions based on the data.
- Wireless communication is used to transmit commands to a remote system.
- The system is capable of sending signals to control the motor's operation.
- The motor's direction and speed are adjusted based on the received instructions.
- The motor's control is precise, ensuring accurate operation according to the processed data.
- The setup is portable, powered by a battery, which eliminates the need for an external power source.

TRANSMITTER SIDE OF ROBOTIC ARM



RECEVIER SIDE OF ROBOTIC ARM



VOLTAGE REGULATOR IC

GRIPPER MECHANISM

12 V DC RECHARGEABLE
BATTERY

3 AXIS ROBOTIC ARM

SERVO MOTOR

ARDUINO NODEMCU

RADIO FREQUENCY(RX)

WORKING PRINCIPLE OF PROPOSED SYSTEM

- **Gesture Recognition:** The system uses flex sensors and a gyroscope to detect hand and finger movements, converting them into electrical signals.
- **Signal Processing:** The Arduino at the transmitter side processes sensor data and converts it into control signals for wireless transmission.
- **Wireless Communication:** RF modules transmit the processed signals from the transmitter (worn by the user) to the receiver (robotic arm side).
- **Robotic Arm Control:** The NodeMCU at the receiver side receives signals and controls servo motors, replicating the user's hand gestures.
- **Security Authentication:** Fingerprint and keypad authentication restrict unauthorized access, ensuring only approved users can operate the robotic arm.
- **Real-Time Feedback:** An LCD screen displays executed commands, movement status, and security authentication results for user confirmation.

ADVANTAGES

- ❖ **Enhanced Safety** – Minimizes human exposure to hazardous environments by enabling remote operation.
- ❖ **Wireless and Gesture-Based Control** – Uses RF communication and hand gestures for intuitive and precise control.
- ❖ **High Precision and Accuracy** – Gyroscope and flex sensors ensure smooth and accurate robotic movements.
- ❖ **Multi-Level Security** – Incorporates fingerprint authentication and keypad access to prevent unauthorized use.
- ❖ **Portability and Energy Efficiency** – Battery-powered design allows for mobility and optimized power consumption.
- ❖ **Versatile Applications** – Can be used in industrial automation, healthcare, and assistive technologies for various tasks.

CONCLUSION

Our project focuses on developing a gesture-controlled wireless robotic arm for safe and efficient operation in hazardous environments. It consists of a transmitter worn by the user, which captures hand gestures using flex sensors and a gyroscope. These sensors detect finger bending and wrist movements, converting them into electrical signals processed by Arduino and transmitted via RF communication. On the receiver side, NodeMCU interprets the signals and controls servo motors to replicate human-like movements. The flex sensor manages the gripper mechanism for precise object handling, such as picking and placing materials. Security is enhanced with fingerprint authentication and a keypad sensor to restrict unauthorized access. An LCD display provides real-time feedback on commands and authentication status, while a voltage regulator IC ensures a stable power supply. The system demonstrates high accuracy in gesture recognition and wireless communication, making it ideal for industrial automation, hazardous material handling, and remote robotic applications.

REFERENCE

- ❖ Singh, Shah, et al. "Design and Development of Robotic Arm for Material Handling Using Hand Glove Controller," 2024 International Conference on Science Technology Engineering and Management (ICSTEM), Coimbatore, India, 2024.
- ❖ Anughna N, Ranjitha V, Tanuja G “Design and Implementation of Wireless Robotic Arm Model using Flex and Gyro Sensor”, International Journal of Recent Technology and Engineering (IJRTE) ISSN: 2277-3878, Volume-8 Issue-5, January 2020.
- ❖ Iyer, S. Tarekar, et al. "Hand Gesture Controlled Robot," 2019 9th International Conference on Emerging Trends in Engineering and Technology - Signal and Information Processing (ICETET-SIP-19), Nagpur, India, 2019.
- ❖ S. S. Dheeban, D. V. Harish, A. Hari Vignesh, and M. Prasanna, “Arduino Controlled Gesture Robot,” Dec. 2018.
- ❖ Priyambada Mishra, Riki Patel, Trushit Upadhyaya, ArpanDesai "Review of Development of Robotic Arm Using Arduino UNO," International Journal on Recent Research in Science, Engineering and Technology, ISSN: 2348-3105 Volume 5, Issue 5, May 2017.

REFERENCE

- ❖ Megalingam, Rajesh Kannan, et al. "Hand Gesture Based Wireless Robotic Arm Control for Agricultural Applications." IOP Conference Series: Materials Science and Engineering. Vol. 225. No. 1. IOP Publishing,(2017).
- ❖ Jayabala, Pradeep. (2016). Design and Implementation of Gesture Controlled Robotic Arm for Industrial Applications. International Journal of Scientific Research. 3. 202-209.
- ❖ Dhepekar, Puja,et al."Wireless robotic hand for remote operations using flex sensor." 2016 International Conference on Automatic Control and Dynamic Optimization Techniques (ICACDOT). IEEE, 2016.
- ❖ Shamsheer Verma “Hand Gestures Remote Controlled Robotic Arm”,Advance in Electronic and Electric Engineering. ISSN 2231-1297, Volume 3, Number 5 (2013),pp. 601-606.
- ❖ Syed, Abidhusain, et al. "Flex sensor based robotic arm controller using micro controller." Journal of Software Engineering and Applications 5.05 (2012).

THANK YOU !